

THE AGENT COLUMN

A higher-resolution zoom into Stratum XIII of the parent atlas — the agent layer unfolded as ten sub-strata plus four wrapping meta-strata.
Stratum agenticum — denso compositum, ordo descendens, gradibus decem cum quattuor strata circumdantia

X SUB-STRATA · IV META · DESCENDING

X	END-USER · SURFACE form factors	chat · cli · ide · trigger · slack · whatsapp · voice · ar	10
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M E T A · W R A P P I N G F O R C E S

A	CAPABILITY · SAFETY	asl · preparedness · frontier · aisi · metr
B	REGULATION	eu ai act · ca sb 53 · sectoral · bartz · nyt
C	ECONOMICS	per-task · outcome pricing · finops
D	GEOPOLITICS	sovereign agents · export controls · jurisdiction

K E Y

sub-stratum

component

meta-stratum



dependency

CAPABILITY · SUBSTRATE

Strata I–III · the model, the runtime, the protocol — the geological basement on which every agent runs.
Stratum primum, secundum, tertium · capacitas, harnesia, protocollum tooluse

III

TOOL · PROTOCOL

MODEL CONTEXT PROTOCOL · ADOPTION · FORK VECTORS

● MCP — Anthropic Nov '24 JSON-RPC; Py + TS SDKs	● OpenAI adoption — Mar '25 Agents SDK · Responses API · mini · Vertex AI · A2A Copilot Studio · SK	● Google adoption — Apr '25 Gemini · Vertex AI · A2A Copilot Studio · SK	● Microsoft adoption — May '25 Copilot · GitHub Copilot · GitHub Copilot Studio · SK
● Hugging Face — Jun '25 MCP-compatible Spaces	● Linux Foundation — Dec '25 Agentic AI Foundation	● Registry · ~11.4K servers Apr '26 LF stats	● Cloudflare Workers AI MCP Dec '24 · gateway
● Kong MCP Gateway GA Jul '25 · F500 logos	● Pomerium identity proxy May '25	● Function calling (legacy) OpenAI Jun '23	● Structured outputs OpenAI Aug '24
● Stripe MCP first-party · canonical	● Linear · Cloudflare MCP first-party tier	● GitHub MCP MSFT canonical	● A2A protocol — Apr '25 agent-to-agent (Google)
● Fork vector — proprietary tool calls OAI latency wins	● Fork vector — quality split long tail abandonware		

stratum tertium · the joint to the world

MCP · JSON-RPC · GATEWAYS

REGISTRY · MAY '26

~11.4K

HYPERSCALERS COMMITTED

4 / 4

GATEWAY CATEGORY

GA

II

RUNTIMES

AGENT RUNTIMES · HARNESSES · ORCHESTRATORS

● Claude Agent SDK Sep '25 · sub-agents · small	● OpenAI Agents SDK Mar '25 · handoffs	● Google ADK Apr '25 · Vertex Agent Ecosystem	● LangGraph 1.0 May '25 · production
● LangSmith obs layer · margin lives	● Mastra TypeScript · WC24 · Vertex	● Pydantic AI Pydantic validation · Logfire	● CrewAI multi-agent · \$18M Oct '24
● Smolagents (HF) minimalist · Dec '24	● AutoGen / AG2 MSR + community fork	● DSPy · BAML compile-time / IR layer	● Inspect AI (UK AISI) eval framework · OSS
● Mistral Agents API May '25 · EU	● Manus AI (CN) Butterfly Effect · viral Q1 2026	● Sarvam AI (IN)	● Aleph Alpha compliance-first · DE

stratum secundum · the harness

SDK · LOOP · SUB-AGENTS

SHIPPED · LAST 14 MO

4 SDKs

LANGGRAPH ARR

\$40–60M (inf)

RUNTIME GROSS MARGIN

THINNEST

I

FOUNDATION · CAPABILITY

AGENTIC CAPABILITY DELTAS · NOT PROVIDER ECONOMICS

● Claude Opus 4.5 Nov '25 · 82.8% SWE-V	● Claude Sonnet 4.5 interactive executor	● GPT-5 / 5.1 thinking ~80% SWE-V w/ tools	● ChatGPT Agent Operator+DR · OSWorld classm
● Gemini 2.5 Pro Deep Think GAIA ~70 · BrowseComp ~50	● Project Mariner Apr '26 · OSWorld ~38	● DeepSeek R1 / R2 open weights · mid-50s SWE-V · open · CN	● Qwen QwQ / Qwen3 SWE-V · open · CN
● Kimi K1.5 / K2 long-context CN · Moonshot	● Meta Llama 4 Apr '25 · -10–15 pts vs #1	● Anthropic sub-agents children · 6mo context degr. past 32K tokens	● Reasoning regression past 32K tokens
● Error compounding ceiling 0.95*20 = 36%	● Computer-use ceiling OSWorld ~50% vs human	● Extended thinking 64K 72% interactive vs back-official	● Tool use in thinking GPT-5 · Deep Think

stratum primum · the agentic delta

TAU · SWE · OSWORLD · BROWSECOMP

SWE-BENCH VERIFIED

82.8%

OSWORLD (MAY '26)

~50%

REASONING-MODEL TIERS

9

COGNITION · LOOPS

Strata IV–V · the inner cognition of the working agent — what it remembers and how it plans the next step.
Stratum quartum et quintum · memoria et ratiocinatio · circuitus interni

V

PLANNING · REASONING

WHAT SURVIVED · TEST-TIME COMPUTE · ECONOMICS

● ReAct (Yao Oct '22) still dominant 2026 loop prompting · alive	● Plan-and-Solve (May '23) lab curiosity · rare prov verifier sub-agent · alive	● Tree-of-Thoughts (May '23) emerging 2025–26	● Reflexion (Mar '23) reasoning model tier opens
● AutoGPT / BabyAGI dead in prod · idea survives	● Planner-executor split adaptable architectural id	● Meta-planning / hierarchical	● o1 / o3 (Sep–Dec '24)
● Claude extended thinking Feb '25 · 64K budgets	● Gemini Deep Think (May '25) super-linear on hard	● DeepSeek R1 / R1-Zero Jan '25 · pure RLVR	● Qwen QwQ-32B · Kimi K1.5 open reasoning
● Tool use IN thinking outer ReAct collapses	● `reasoning_effort` knob OpenAI · buyer-controlled	● PRM (Lightman May '23) n-model · not exposed	● GRPO at trajectory DeepSeek · default RL

stratum quintum · the inner loop

REACT · PLANNER-EXECUTOR · TTC

CX TRAJECTORY · TOKENS

4–25K

CODE AGENT · TOKENS

30–250K

DEEP RESEARCH · TOKENS

0.5–5M

IV

MEMORY · STATE

STANDALONE OR ABSORBED · COMPLIANCE WEDGE

● Memo0 (YC W24) \$24M A Dec '25 · 40K state \$10M seed · A rumored '26	● Letta (MemGPT) \$15M A Feb '25 · HIPAA	● Zep — Graphiti OSS pipeline · seed Dec '24	● Cognee Sep '25 paper · CN
● Microsoft GraphRAG Apr '24 paper · Jul '24 @ \$5M '24 · cheaper graph	● LightRAG (HKU) research-only	● HippoRAG · PathRAG · OG-RAG	● MemoryOS (Tsinghua)
● ChatGPT memory Apr '24 paid · Sep '24 free	● Claude Projects memory Apr '26	● Gemini personalization GA Mar '26	● Working memory (LangGraph) engineering not category
● Procurement — Article 17 Mem0 forget API Feb '26 Athelas · F500 health	● HIPAA story (Zep)	● Contextual integrity Nissenbaum · no vendor	● Falsifiability test 70% pitches = RAG + writes

stratum quartum · persistence

EPISODIC · SEMANTIC · PROCEDURAL

STANDALONE CEILING (INF) LAB MEMORY MAU

\$200–400M

~800M

CRUX #5 (INHERITED)

OPEN

ACTION · DEFENSE

Strata VI–VIII · where the agent reaches into the world, how we observe that it did the right thing, and what defends when it does not.
Stratum sextum, septimum, octavum · operatio, observatio, custodia

VIII

RUNTIME · SAFETY

GUARDRAILS · INJECTION DEFENSE · BUNDLING

● Lakera (Guard · Red) Series A '23 · B Q4 '25	● Protect AI · to Palo Alto ~\$700M Q3 '25 · the bell Aug '24 · AI Defense	● Robust Intelligence · to Cisco \$50M B Mar '24	● HiddenLayer
● Apex Security Sequoia seed Mar '24	● CalypsoAI \$23M B · federal angle	● Cranium KPMG spin · GRC-leaning Apr '24 · Auto Reasoning	● AWS Bedrock Guardrails
● Purview AI Hub · Defender MSFT Ignite Nov '24	● Vertex AI Model Armor Cloud Next '24	● NeMo Guardrails 1.0 GA Apr '26 · OSS	● Llama Guard 3 · Prompt Guard 2 Meta open weights
● Promptfoo (now OAI) red-team OSS · Sep '25	● Garak (NVIDIA OSS) vulnerability scanner	● EchoLeak CVE class Mar '26 · indirect PI	● Constitutional classifiers Anthropic · baked in

stratum octavum · the tool boundary

PI · ACTION GATES · SANDBOXING

PURE · PLAYS ABSORBED

18 MO

INDIRECT PI

UNSOLVED

DEFENSE STACK · LAYERS

5

VII

EVAL · OBSERVABILITY

AGENT EVAL · OBS · BENCHMARKS

● Braintrust \$36M A · B rumored '26	● LangSmith / LangChain \$25M C · LangGraph 1.0	● Langfuse OSS · YC W23 · EU choice	● Arize Phoenix / AX \$70M C Dec '24
● Galileo \$45M B · C reported Apr	● HumanLoop U26 · FSI	● Patronus AI · Lynx · Glider \$17M A · defensible judge	● Comet Opik · DeepEval @SS · pytest-style
● AgentOps CrewAI / AutoGen ecosystem	● Helicone · proxy assessing as native matures	● Promptfoo (acq OAI) Sep '25 · neutrality gone	● Inspect AI (UK AISI) EU AI Act conformity
● METR · time-horizon Mar '25 · cite not procured	● TAU-bench / tau-2 Sderra-co · CX proxy	● SWE-Bench Verified · Live code agency · ~75-80%	● OSWorld · WebArena · GAIA computer/web/research

stratum septimum · trajectory analytics

TRAJECTORY · REPLAY · ONLINE

OTEL GENAI CONVENTIONS

JAN '26 GA

CONSOLIDATION (2027)

PARTIAL

PROCUREMENT GAP

OPEN

VI

ACTION · SURFACES

SANDBOXED CODE · BROWSER · COMPUTER · VOICE OUTBOUND

● E2B OSS-core · ~\$15-25M ARR	● Modal agent-workload pivot · >\$50M	● Daytona price-aggressive GA Jan '26 · Firecracker	● Vercel Sandbox
● Replit Agent sandbox bundled · prosumer	● HF Spaces · Codespaces ad-hoc · weaker fit	● Browserbase + Stagehand >\$50M ARR Q1 '26	● Browserless · Skyvern incumbent · OSS
● Playwright + AI the boring baseline	● Anchor · Hyperbrowser MCP-compat 2025	● Anthropic Computer Use Oct '24 · Claude for Chrome rebrand mid '25	● ChatGPT Agent
● Google Project Mariner GA Apr '26 · Gemini Ent	● MSFT Copilot Vision Windows-native	● Vapi · Retell · Bland voice trio · \$20M+ ARR	● LiveKit Agents eWebRTC substrate
● ElevenLabs Conv.AI voice up-stack	● Cartesia Sonic-2 ~40ms first-byte	● Deepgram Nova-3 ASR · accent	● Twilio · Telnyx · LiveKit Cloud telephony

stratum sextum · agent reaches into world

SANDBOX · BROWSER · COMPUTER · VOICE

VOICE · TO · VOICE FLOOR

500-800ms

SANDBOX TIMELINE

18-24 MO

CU PRODUCTION

NO

PRODUCTIZED · LAYER

Strata IX-X - where enterprise revenue lands and where the human meets the agent – the productized application and its surface.
Stratum nonum et decimum - productum agenticum et facies hominis

X

END-USER · SURFACES

TWELVE FORM FACTORS · THE BINDING CONSTRAINT

- | | | | |
|---|---------------------------|---------------------------|------------------------------|
| ● Chat — web/desktop | ● Chat — mobile/on-device | ● CLI / terminal | ● In-IDE inline |
| ChatGPT 800M WAU · plate apple Intelligence slot Claude Code ~\$500M RR | | | Cursor · Copilot · Windsurf |
| ● Agentic trigger | ● In-browser sidebar | ● Embedded in SaaS | ● Slack / Teams · headless |
| n8n · Zapier · cron · ArCloud for Chrome · ComeComPilot 365 ~30M seats | | | Glean wedge |
| ● WhatsApp · SMS · email | ● Voice — inbound | ● Wearable · AR · ambient | ● Computer use (UX) |
| Meta API Jan '26 · 2.9B Advanced Voice · AirPodsRay-Ban >2M · Plaud >1M | | | watched only · no reg ent |
| ● API · SDK · no-UI | ● WhatsApp Business API | ● Coze on Feishu · WeChat | ● Manus · Yellow.ai · Haptik |
| 2/3 OAI rev · majority AdmIrdBSc · GLOBAL MAJORITYCN parallel-track | | | non-US agent surfaces |
| ● Pew chat-churn 41% | ● Anthropic mobile gap | | |
| Jan '26 · 'chat solved' noSMAI slot · no voice | | | |

stratum decimum · form factor

B U Y E R . T H R E A T . A D O P T I O N

\$5B+

2.9B

FORM

IX

VERTICAL · PRODUCTS

ENTERPRISE REVENUE LAYER · BUYING MOTION BY DOMAIN

- | | | | |
|---|--|--|--|
| ● Sierra —\$175M+ ARR
\$10B (Mar '26) · NYC dual | ● Decagon —\$80M+
\$4.5B Sep '25 · NYC/SF | ● Glean —\$300M+
\$7.2B Sep '25 · NYC growth | ● Harvey —\$100M+
\$15B Feb '26 · NYC office |
| ● Hippocratic —\$50M+
\$2B Feb '26 · NYC HQ | ● Augment Code —\$40M
\$977M Nov '25 · enterprise | ● Cursor —\$500M RR
\$9.9B Jul '25 · IDE lead | ● Cognition (Devin)
\$5B Codeium absorbed Mar '26 |
| ● Replit Agent —\$150M
\$3B Sep '25 · prosumer | ● Lovable —\$80M Q1'26
fastest EU SaaS · Stockholm | ● Abridge —\$120M+
\$2m 75B Jun '25 · scribe | ● Hebbia —\$50M+
NYC HQ · ex-2Sigma · raise |
| ● Clay —\$80M+
NYC HQ · \$1.5B Jan '26 | ● Rogo —\$30M+
NYC HQ · \$400M Jan '26 | ● Ramp AI org
13B NYC 20B raise rumor | ● Runway —\$100M+
NYC Chelsea · creative |
| ● ElevenLabs —\$200M+
voice platform · \$3.3B | ● Cresta · Ada · Intercom Fin
CX maturing/stalling | ● Parloa · PolyAI · Regal
EU/UK + NYC sales voice | ● EvenUp · Eve · Spellbook
legal long tail |
| ● 11x · Artisan · AiSDR
synthetic SDR · 11x flat | ● Mistral Le Chat Ent.
€600M Mar '26 · EU sovereign | | |

stratum nonum · productized agents

C X . C O D E . L E G A L . H E A L T H . F I N A N C E

3 / 9

10+

CX ONLY

S U B S T R A T E . V O L I I I . T H E A G E N T L A Y E R

Compiled for A. Yedi · Cycle MMXXVI · Rev. I

PLATE V OF VI

META · WRAPPING FORCES

Meta A–D · the four forces that wrap the agent layer: capability–safety regimes, regulation, economics, geopolitics.
Quattuor stratorum meta · circumscribunt agentcam stratam

D

GEOPOLITICS

WHERE AGENTS RUN · EXPORT CONTROLS · SIT-CONTROLS

- Mistral Le Chat Ent. €600M @ €11B Mar '26 Humain ties · sovereign AI 300/350 commits · \$10B \$41M · Q1 '26 raise rumor
- Krutrim (Ola IN) \$50M · political tailwind Butterfly · \$500M rumor ByteDance · parallel-track agent Q2 '26
- US Commerce auth — Nov '25 G42 + Humain chips 2026–27 regulatory front AI Act + GDPR + Schrems Where memory + actions reside
- G42 · Falcon · UAE
- Manus (CN)
- Coze on Feishu
- WeChat Mini-Programs
- Humain (KSA)
- Sarvam AI (IN)
- Agent-capability export rule
- EU residency lever
- Trajectory siting

meta · sovereign agents

S O V E R E I G N · E X P O R T · J U R I S D I C T I O N

C

ECONOMICS

FROM PER-TOKEN TO PER-TASK · OUTCOME PRICING

- Token price · –4× / –10× Q1 '24 to Q1 '26
- CX trajectory · 4–25K tok ~ 0.02–0.30 / task
- Code agent · 30–250K tok ~ 0.20–3 / task
- Deep research · 0.5–5M tok ~ 1–20 / task
- Planner-executor split 70–85% on cheap tier
- Vercel AI Gateway routing first-class · GA '26
- AWS Bedrock · OpenRouter tier routing standard
- Outcome pricing — CX per-resolution · \$1–4 (Taylor)
- Outcome — healthcare per-encounter · per-call \$30/hr knowledge · financial –18 mo window
- FinOps for Tokens (Bet #4)
- Hyperscaler bundling Bedrock auto-optim threat
- Per-seat survivors

meta · the inversion

P E R - T A S K · O U T C O M E · F I N O P S

B

REGULATION

EU ARTICLE 14 · CA SB 53 · BARTZ · SECTORAL

- EU AI Act · Art. 55 GPAI enforceable Aug 2 '25
- EU AI Act · Art. 14 oversight draft Apr '26 · agent Q1 '26 · Inspect AI cited effective Jan 1 '26
- EU AI Act · conformity drafts
- CA SB 53 frontier disclosure
- NY SS641 · state-level advancing 2026
- Trump Dec '25 preemption EO courts through '26–'27
- Bartz v. Anthropic \$1.5B training-data settlement vendor terms repricing
- NYT v. OpenAI
- FINRA · OCC · BD rules finance agent fence
- HIPAA · FDA SaMD health agent fence
- TCPA · state consent voice agent fence
- EU AI Office structural regulator

meta · the gauntlet

E U A I A C T · C A S B 5 3 · S E C T O R A L

A

CAPABILITY · SAFETY

LAB-SIDE SAFETY REGIMES · EVAL INSTITUTES

- Anthropic ASL framework ASL-3 Opus 4 May '25
- Anthropic RSP — Nov '24 binding policy
- OpenAI Preparedness Framework tracked categories
- Preparedness adjustment clause Risk #3 · open question
- Google Frontier Safety May '24 · maturing
- DeepMind capability evals auto-replication · cyber Nov '23 · Inspect framework early '26 · narrower scope
- UK AISI
- US AISI (post-restruct)
- METR (research org) time-horizon · 7-mo double-blind regulatory counterpart 2026–27 question
- EU AI Office
- ASL-4 capability emergence
- Model cards for agents in-flight standardization

meta · the deploy gate

A S L · P R E P A R E D N E S S · F R O N T I E R